

Learning Objective 2.2: I can simulate a dipole antenna using an EM simulation tool and interpret the results against analytical predictions.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Segmentation arithmetic.** You are about to model a straight center-fed dipole at $f = 2.45\text{GHz}$ in NEC. The wire is 0.8 mm in diameter, so the radius is $a = 0.4$ mm. Before touching the keyboard, size the model on paper.

- (a) Find the free-space wavelength and the physical length of a half-wave dipole at this frequency.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{2.45 \times 10^9 \text{ Hz}} = 0.1224 \text{ m} = 122.4 \text{ mm}$$

$$\frac{\lambda}{2} = 61.2 \text{ mm}$$

$$\lambda = \boxed{122.4 \text{ mm}}$$

$$\lambda/2 = \boxed{61.2 \text{ mm}}$$

- (b) You choose $N = 15$ segments. Find the segment length Δ and show that it satisfies both NEC bounds: $\Delta < \lambda/20$ and $\Delta > 8a$.

$$\Delta = \frac{61.2 \text{ mm}}{15} = 4.08 \text{ mm} = 0.033 \lambda$$

$$\Delta = \boxed{4.08 \text{ mm}}$$

Upper bound: $\lambda/20 = 6.12 \text{ mm}$, and $4.08 < 6.12$, so the segment is short enough that phase barely changes across it. Lower bound: $8a = 8(0.4) = 3.2 \text{ mm}$, and $4.08 > 3.2$, so the thin-wire kernel is still valid. Both bounds pass, and 15 segments over a half wavelength sits inside the 10–20 rule of thumb.

- (c) What is the largest *odd* segment count this wire will tolerate before it violates the $\Delta > 8a$ rule?

$$\Delta_{min} = 8a = 3.2 \text{ mm}$$

$$N_{max} = \frac{61.2 \text{ mm}}{3.2 \text{ mm}} = 19.1 \rightarrow N = 19$$

$$N_{max} = \boxed{19}$$

This is a fat wire: the refinement ceiling sits barely above the segmentation you already chose. If a convergence study demanded 41 segments, this model could not supply them — the answer would be to use the extended thin-wire kernel or a thinner wire, not more segments.

- (d) In one or two sentences, explain why the segment count must be odd.

The voltage source is applied to a single segment, and that segment is the antenna's feed point. An odd count puts one segment exactly at the center of the wire, so the feed is symmetric. With an even count the center falls on a junction between two segments and the source must be placed on one side of center, which breaks the symmetry of the model and shifts the reported input impedance.

2. **Predict, then account for the difference.** You model the 2.45 GHz dipole of Question 1 at *exactly* $\lambda/2 = 61.2$ mm, with 15 segments and a 1 V source on the center segment. NEC returns $Z_{in} = 86.9 + j46.2\Omega$ and a peak gain of 2.17 dBi.

- (a) State the four predictions Lesson 7 makes for a half-wave dipole: input impedance at exactly $\lambda/2$, resonant length, input resistance at resonance, and directivity in dBi.

From the sinusoidal-current analysis: $Z_{in} \approx 73 + j42.5\Omega$ at exactly $\lambda/2$; the antenna is resonant slightly short, at about 0.47 to 0.48 λ ; the input resistance there is about 70Ω ; and the directivity is $D = 1.64$, or 2.15 dBi, with a half-power beamwidth of 78° .

- (b) Compute the percent difference between the simulated and predicted values of R_{in} and of X_{in} .

$$\frac{86.9 - 73}{73} = 0.190 \rightarrow 19.0\%$$

$$\frac{46.2 - 42.5}{42.5} = 0.087 \rightarrow 8.7\%$$

$$R_{in} \text{ error} = 19.0\%$$

$$X_{in} \text{ error} = 8.7\%$$

The reactance is close; the resistance is not. Note also that the simulated gain, 2.17 dBi against a predicted 2.15 dBi, is off by 1% — the pattern quantities agree even though the impedance does not.

- (c) In two or three sentences, account for the resistance difference. Be specific about the mechanism — “simulation error” earns nothing.

The value $73 + j42.5\Omega$ is the impedance of an assumed sinusoidal current, and a sinusoid resonates when the wire is about 0.486λ long. A real wire has finite radius, stores energy in the near field around that radius, and resonates shorter — near 0.47λ . A wire cut to exactly $\lambda/2$ is therefore already about 5% long: it is inductive, and its resistance has already climbed well up the steep part of the R_{in} versus length curve. The two numbers describe two different antennas, not one antenna and one error.

- (d) Trimming brings this wire to resonance at 0.473λ . Find the resonant length in millimetres and the percent shortening from $\lambda/2$.

$$L_{res} = 57.9 \text{ mm}$$

$$L_{res} = 0.473(122.4 \text{ mm}) = 57.9 \text{ mm}$$

$$\text{shortening} = \frac{61.2 - 57.9}{61.2} = 0.054 \rightarrow 5.4\%$$

$$\text{trim} = 5.4\%$$

Roughly a 5% trim, which is the standard rule of thumb for bringing a dipole to resonance.

3. **Reading a convergence study.** A classmate models a dipole of length exactly 0.5λ with wire radius $a = 0.001 \lambda$ and re-runs it at five segment counts:

Segments N	$R_{in} (\Omega)$	$X_{in} (\Omega)$
5	79.9	+35.9
11	81.9	+43.9
21	83.4	+45.6
41	84.5	+46.5
81	85.4	+47.2

- (a) Compute the percent change in R_{in} from each row to the next.

$$\begin{aligned}
 5 \rightarrow 11 : \quad & \frac{81.9 - 79.9}{79.9} = 2.5\% \\
 11 \rightarrow 21 : \quad & \frac{83.4 - 81.9}{81.9} = 1.8\% \\
 21 \rightarrow 41 : \quad & \frac{84.5 - 83.4}{83.4} = 1.3\% \\
 41 \rightarrow 81 : \quad & \frac{85.4 - 84.5}{84.5} = 1.1\%
 \end{aligned}$$

last step =

1.1%

The steps shrink, but slowly: R_{in} is still creeping up by about a percent per doubling.

- (b) Find the largest segment count the $\Delta > 8a$ rule permits for this wire. Which rows of the table violate it?

$$\begin{aligned}
 \Delta &= \frac{0.5 \lambda}{N} > 8a = 0.008 \lambda \\
 N &< \frac{0.5}{0.008} = 62.5 \rightarrow N_{max} = 61
 \end{aligned}$$

$N_{max} =$

61

The $N = 81$ row breaks the rule — its segments are only about five radii long — so it sits outside the range where the standard thin-wire kernel is trustworthy and should not be used as evidence of anything.

- (c) Which segment count would you defend for this model, and why? Your answer must reference both the convergence trend and the kernel limit.

41 segments. Each further refinement moves R_{in} by only about one percent, well below what you could measure on the bench, and $\Delta = 0.0122 \lambda$ clears the 0.008λ floor with room to spare for a shorter trimmed wire. Going to 81 buys a percent of drift at the cost of leaving the kernel's valid range.

- (d) Explain in one or two sentences why gain converges at a lower segment count than input impedance does.

Gain is an integral over the entire current, so errors in individual segments partly cancel and the total settles quickly. Input impedance is read from one segment — the one carrying the source — so it inherits the full local error of the feed model, which itself changes every time the segment length changes.

4. **The average-gain audit.** After a full-sphere pattern run of a lossless dipole in free space, NEC reports an average power gain of 1.15.

- (a) What does an average power gain of exactly 1.000 tell you about a lossless free-space model, and why is that a test rather than a result?

Average power gain is the gain averaged over the whole sphere, which is the total radiated power divided by the total input power. For a lossless antenna in free space, every watt delivered to the terminals must leave as radiation, so the average must be 1.000. It is a test because you already know the answer: any deviation is not physics, it is a defect in the model, and it is the cheapest error check available.

- (b) Express the reported 1.15 in decibels, and state whether you would accept the run.

$$10 \log_{10} (1.15) = 0.61 \text{ dB}$$

avg gain =

0.61 dB

Reject it. The model claims to radiate 15% more power than it was fed, which is impossible; a healthy run lands inside roughly $\pm 5\%$, or about ± 0.2 dB. Nothing else in this output file — impedance, gain, pattern — can be trusted until the cause is found.

- (c) Give two plausible causes of an average gain this far from 1.000, and the specific check that would confirm each.

First, a segmentation violation: segments shorter than about eight wire radii break the thin-wire kernel and corrupt the currents. Check it by computing Δ and $8a$ by hand, and re-run with fewer segments or a thinner wire. Second, a geometry or excitation error — wires that do not actually connect, a duplicated wire, or a source placed on a segment that is not at the feed. Check it by plotting the geometry and the segment numbering, and confirming the source tag and segment index against the wire card. A third common cause is requesting the average over less than a full sphere, which makes the number meaningless rather than wrong.

5. **Why the current comes first.** A classmate claims that NEC “computes the radiation pattern directly from the geometry.”

- (a) In two or three sentences, explain why a method-of-moments code must solve for the current distribution before it can produce a pattern. Refer to the radiation integral.

The far field is the radiation integral of the current over the antenna: the geometry alone tells you where the current can flow, not how much flows where or with what phase. The method of moments exists precisely to supply that missing current — it discretizes the wire, enforces $E_{tan} = 0$ on the conductor, and solves a linear system for the segment currents. Only then does it evaluate the same radiation integral you would evaluate by hand, so the geometry sets up the problem and the current answers it.

- (b) The solver drives the center segment with 1 V and reports a feed current of $8.90 - j4.80$ mA. Find Z_{in} , then the magnitude of the reflection coefficient and the VSWR referred to 50Ω .

$$Z_{in} = 87.0 + j46.9\Omega$$

$$\begin{aligned} Z_{in} &= \frac{V}{I} = \frac{1}{(8.90 - j4.80) \times 10^{-3}} \\ &= \frac{(8.90 + j4.80) \times 10^{-3}}{1.0225 \times 10^{-4}} = 87.0 + j46.9\Omega \end{aligned}$$

$$|\Gamma| = 0.412$$

$$\Gamma = \frac{Z_{in} - 50}{Z_{in} + 50} = \frac{37.0 + j46.9}{137.0 + j46.9}$$

$$\text{VSWR} = 2.40$$

$$|\Gamma| = \frac{59.7}{144.8} = 0.412$$

$$\text{VSWR} = \frac{1 + 0.412}{1 - 0.412} = 2.40$$

- (c) You need this antenna matched to 50Ω . Would you trim the wire to resonance or add a matching network? Give the trade-off in one or two sentences.

Trim first. Trimming costs nothing, removes the $+j46.9\Omega$ entirely, and leaves about 72Ω real, which is already a VSWR near 1.4 on a 50Ω line — acceptable for most links. A matching network would bring the residual mismatch closer to unity but adds parts, loss, and a bandwidth restriction, so it is worth doing only when the remaining mismatch actually costs you something in the link budget.

Documentation: _____